Anti-carcinogenic effects of sulforaphane in association with its apoptosis-inducing and anti-inflammatory properties in human cervical cancer cells.

BACKGROUND: The multistep process of carcinogenesis is characterized by progressive disorganization and occurrence of initiation, promotion, and progression events. Several new strategies such as chemoprevention are being developed for treatment and prevention at various stages of carcinogenesis. Sulforaphane, a potential chemopreventive agent, possesses anti-proliferative, anti-inflammatory, anti-oxidant and anti-cancer activities and has attracted extensive interest for better cancer management. METHODS: We evaluated the effect of sulforaphane alone or in combination with gemcitabine on HeLa cells by cell viability assay and confirmed the results by apoptosis assay. Further we analyzed the effect of sulforaphane on the expression of Bcl-2, COX-2 and IL-1β by RT-PCR on HeLa cells. RESULTS: In the present study, sulforaphane was found to induce dose-dependent selective cytotoxicity in HeLa cells in comparison to normal cells pointing to its safe cytotoxicity profile. Additionally, a combination of sulforaphane and gemcitabine was found to increase the growth inhibition in a synergistic manner in HeLa cells compared to the individual drugs. Also, the expression analysis of genes involved in apoptosis and inflammation revealed significant downregulation of Bcl-2, COX-2 and IL-1β upon treatment with sulforaphane. CONCLUSION: Our results suggest that sulforaphane exerts its anticancer activities via apoptosis induction and anti-inflammatory properties and provides the first evidence demonstrating synergism between sulforaphane and gemcitabine which may enhance the therapeutic index of prevention and/or treatment of cervical cancer.